

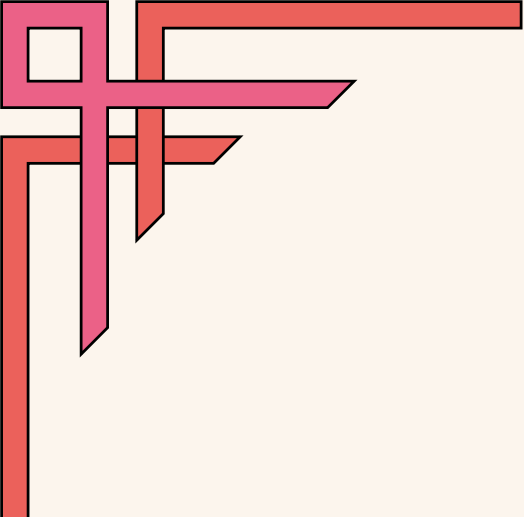
Week-1 Deliverable

**TICKET MAPPING
REPORT**

Team-2

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1. Introduction

The Customer Insights and Data Analytics Early Remote Internship aims to acquire practical abilities in evaluating customer support data to improve service operations and customer experience. During this internship, we will work together to learn how structured data analysis can enable efficient issue resolution, informed decision-making, and continual service improvement.

The Week 1 "**Ticket Mapping Report**" activity focuses on establishing a solid foundation in customer support operations by examining a dataset named "***Freshdesk Customer Support Tickets Dataset***". This activity involves carefully reviewing and categorizing support tickets based on issue type, priority, status, tags, and customer sentiment. This organized mapping helps us understand how customer issues are sorted, prioritized, and monitored across the support lifecycle.

Furthermore, we examine the mapped tickets to find reoccurring issues and typical consumer complaints. By identifying trends across several tickets and replicating an entire end-to-end support workflow for a certain reoccurring issue, we obtain insight into successful ticket triage, SLA determination, and resolution processes. This study highlights how our team approaches data-driven analysis to improve support efficiency, operational consistency, and overall customer happiness.

2. Objectives

Our aim is -

- To build a foundational understanding of customer support operations through structured ticket analysis.
- To categorize support tickets based on issue type, priority, status, tags, and customer sentiment.
- To identify recurring customer issues and common pain points through pattern analysis.
- To simulate an end-to-end support workflow for a selected recurring issue, including prioritization and SLA definition.
- To develop data-driven insights that support improved service efficiency and customer satisfaction.

3. Dataset Overview

We analyzed the assigned dataset "***Freshdesk customer support tickets dataset***" containing **30 support tickets**. The dataset captures real-world customer interactions handled by the support team over a defined time period and provides essential information related to ticket

creation, resolution, and issue description. Each record represents a unique customer issue managed by an assigned support agent.

Number of rows (tickets): 30 (Ticket IDs T011 to T040)

Number of columns: 11

So the dataset is **30 × 11** in size.

Column Name	Data Type	Description and Analytical Importance
Ticket ID	Categorical (Identifier)	Unique identifier for each support ticket; enables tracking, referencing, and analysis of individual issues.
Created Date	Date & Time	Records when the ticket was raised by the customer; essential for measuring response times and SLA compliance.
Resolved Date	Date & Time	Indicates when the issue was resolved; used to calculate resolution time and assess support efficiency.
Agent	Categorical	Identifies the support agent assigned to the ticket; helps evaluate agent workload and productivity.
Customer	Categorical	Represents the customer who submitted the ticket; useful for customer-level analysis and interaction tracking.
Subject / Short Description	Text	Summarizes the customer issue; serves as the primary input for categorization, tagging, and sentiment analysis.
Category	Categorical	Classifies the type of issue (e.g., Technical, Billing, Feature Request); helps identify common problem areas.
Priority	Categorical	Indicates urgency level of the issue; guides SLA timelines and resolution sequencing.
Status	Categorical	Reflects the current stage of the ticket (Open, Pending, Resolved, Closed); supports workflow tracking.

Tags	Categorical (Multi-label)	Provides thematic labels for grouping similar issues; enables pattern recognition and faster ticket routing.
Sentiment	Categorical	Captures customer emotion (Positive, Neutral, Negative); helps assess customer experience impact and urgency.

This organized dataset serves as a solid foundation for analyzing customer support tickets, and the originally empty columns- **Category**, **Priority**, **Status**, **Tags**, and **Sentiment** ; which will be filled throughout the ticket mapping process to facilitate pattern recognition and workflow modeling.

4. Methodology

This section outlines the standardized approach used to review, classify, and analyze customer support tickets on an ongoing basis. The methodology focuses on consistent application of decision rules rather than reporting outcomes.

4.1 Ticket Review Process

Each support ticket is reviewed individually using the ticket subject and short description as the primary sources of information. These fields are examined to understand the customer issue, particularly when other fields are incomplete or missing. Tickets are reviewed sequentially to ensure a consistent interpretation across the dataset.

4.2 Categorization Logic

When fields such as Category, Priority, Status, Tags, and Sentiment are missing, they are filled in manually based on the information provided in the ticket subject and description.

- Category is chosen based on the main issue described.
- Priority is set based on how urgent or serious the issue appears.
- Status reflects whether the issue seems unresolved or completed.
- Tags are added as short keywords to summarize the issue.
- Sentiment is identified as positive, neutral, or negative based on the tone of the message.

The same rules are used for all tickets to keep the classifications consistent.

4.3 Pattern Identification Approach

After reviewing individual tickets, the full list is checked to find common issues and repeated themes. Similar tickets are compared to make sure they are classified in the same way. This helps identify frequent problems and improve consistency over time.

5. Part 1: Ticket Categorization

5.1 Categorization Criteria

How Categories Are Decided

★ Issue type

- *Technical*: this includes issues that involve any issues related to bugs, errors, connectivity etc.
- *Billing*: issues related to subscription and payment.
- *Feature Request*: tickets related to suggestions for features to be added.
- *Account Management*: tickets related to making account changes

★ Status

- *Open*: no action has been taken yet
- *Pending*: the issue is being identified to find a solution
- *Resolved*: the problem has been fixed, confirmation from customer may be needed
- *Closed*: no more further action is going to be taken

★ Priority

- *Low*: assigned to tickets that are not problems and do not **require** action
- *Medium*: assigned to tickets that require action but not immediately
- *High*: assigned to tickets that are that require quick action
- *Urgent*: assigned to tickets that must be resolved immediately as they could be detrimental

★ Sentiment

- *Positive*: the customer seems to be happy and does not have any complaints
- *Neutral*: the customer does not seem to share any particular sentiment
- *Negative*: customer may be frustrated or unsatisfied

★ Tags

- *Login-issue*: customer struggling to access account
- *Payment-failed*: the payment did not go through

- *Incorrect-billing*: incorrect amount of money was charged
- *Refund*: related to refunds
- *Server-error*: technical issues that are caused by issues from the server
- *Connection-error*: issues caused by a bad connection between customer and server
- *Account-changes*: editing account details
- *Feature*: suggestions for usable features
- *Design*: visual suggestions

5.2 Ticket Categorization Table

The table below presents the complete categorization of **30 customer support tickets**, with all analytical fields systematically populated using the defined criteria. Each ticket has been mapped consistently across category, priority, status, tags, and sentiment to support further analysis in subsequent sections.

The completed ticket categorization table is presented below:

Ticket ID	Subject/Short Description	Category	Priority	Status	Tags	Sentiment
T011	Account locked after multiple attempts	Technical	High	Closed	login-issue	Negative
T012	Subscription auto-renewal failed	Billing	Urgent	Resolved	payment-failed	Neutral
T013	Error during checkout	Technical	Urgent	Resolved	server-error	Negative
T014	Request for multiple language support	Feature Request	Low	Closed	feature	Positive
T015	Cannot change profile picture	Account Management	Medium	Resolved	account-changes	Neutral
T016	Invoice PDF not downloadable	Technical	High	Resolved	server-error	Negative
T017	Password reset link expired	Account Management	High	Closed	login-issue	Neutral
T018	App crashes on settings page	Technical	Medium	Resolved	server-error	Negative
T019	Refund not received after cancellation	Billing	Medium	Resolved	refund	Negative
T020	Add dark mode for dashboard	Feature Request	Low	Closed	design	Positive
T021	Change of email not reflecting	Account Management	High	Resolved	account-changes	Negative
T022	Two-factor authentication not working	Technical	High	Resolved	server-error	Negative
T023	Incorrect tax applied	Billing	Urgent	Closed	incorrect-billing	Negative

	on invoice					
T024	Add custom notification sounds	Feature Request	Low	Closed	feature	Positive
T025	Payment gateway timeout error	Technical	Urgent	Resolved	connection-error	Negative
T026	Unable to link social media account	Account Management	Medium	Resolved	account-changes	Neutral
T027	Locked out after password reset attempt	Technical	High	Closed	login-issue	Negative
T028	App slow when uploading images	Technical	Medium	Resolved	connection-error	Negative
T029	Request for weekly activity summary email	Feature Request	Low	Closed	feature	Positive
T030	Charged extra for subscription plan	Billing	Urgent	Resolved	incorrect-billing	Negative
T031	Cannot update contact information	Account Management	High	Resolved	account-changes	Neutral
T032	Login blocked due to suspicious activity	Technical	High	Closed	login-issue	Neutral
T033	Error 404 when accessing profile	Technical	High	Resolved	server-error	Negative
T034	Add option to export data	Feature Request	Low	Closed	feature	Positive
T035	Payment failed multiple times	Billing	Urgent	Resolved	payment-failed	Negative
T036	Issue with account verification	Account Management	High	Closed	server-error	Negative
T037	Cannot log in after password reset	Technical	High	Resolved	login-issue	Negative
T038	App freezes during checkout	Technical	Urgent	Resolved	server-error	Negative
T039	Request for light theme in dashboard	Feature Request	Low	Closed	design	Positive
T040	Refund request delayed	Billing	Medium	Resolved	refund	Negative

To view the updated Dataset, click here [↓](#):

[+ Freshdesk Customer Support Tickets Dataset Filled](#)

The categorization values were inferred primarily from the **Subject/Short Description** column, supported by **Created Date** and **Resolved Date** to understand urgency and resolution context. This structured mapping enables consistent issue classification, priority handling, trend

identification, and supports more effective customer support analysis and decision-making in subsequent sections.

6. Part 2: Recurring Issues Identification

6.1 Pattern Analysis

Recurring issues are identified by reviewing all categorized tickets and grouping them based on Category, Tags, and similar subject descriptions. A simple counting approach is used to observe how often certain issue types and tags appear across the 30 tickets.

By comparing tickets with the same category (e.g., Technical, Billing) and repeated tags such as login-issue, server-error, and payment-failed, clear patterns emerge. Issues that appear multiple times across different ticket IDs are treated as recurring. This grouping helps highlight common customer problems and areas that require consistent support attention.

6.2 Top 3 Recurring Issues

1. Login and Account Access Issues

Several tickets relate to users being locked out, failed password resets, or login blocks. This suggests challenges with authentication flows and security controls that affect user access.

2. Technical Errors and App Stability Issues

Multiple tickets report server errors, app crashes, slow performance, or checkout failures. These issues point to system stability limitations and the need for technical optimization.

3. Billing and Payment Problems

Repeated cases of payment failures, incorrect charges, and delayed refunds appear across billing tickets. This indicates potential issues with the payment gateway or billing process clarity for users.

7. Part 3: Support Flow Simulation & Diagram

This section simulates a complete end-to-end customer support workflow for a selected recurring issue identified in the dataset. The purpose of this simulation is to demonstrate how a commonly occurring customer problem is handled systematically from ticket creation through final resolution and closure, based on realistic support operations.

7.1 Selected Issue Overview

The recurring issue selected for the support flow simulation is **Login and Account Access Issues**.

This issue category includes customer problems such as account lockouts, failed login attempts, expired password reset links, and login blocks triggered by security mechanisms. In the analyzed dataset, several tickets were tagged consistently as **login-issue**, indicating that access-related problems occur frequently across different users and dates.

This issue was selected because of both its **high frequency** and **high impact**. Login and access issues directly prevent customers from using the platform, making them critical to resolve quickly in order to maintain service continuity and customer satisfaction.

7.2 Support Handling Configuration

To simulate realistic support operations based on the dataset, the following configuration is applied to the selected recurring issue.

Assigned Tags

- **login-issue**

This tag is consistently used in the dataset to represent all login and account access problems, including authentication failures and account lockouts.

Priority Level

- **High Priority**

Login and account access issues are classified as high priority because they block customers from accessing essential platform features, requiring prompt attention from the support team.

SLA Timelines

The SLA timelines are aligned with the assigned priority level to ensure timely handling.

- **First Response Time:** Within **1 hour**
- **Resolution Time:** Within **4–6 hours**

These timelines reflect realistic support expectations for high-priority access-related issues.

7.3 End-to-End Support Workflow

The following workflow outlines the complete support handling process for a login and account access issue.

1. Ticket Creation

- The customer submits a support ticket describing the login or access problem.
- The ticket is logged in the system with a unique ticket ID and timestamp.
- Initial ticket status is set to **Open**.

2. Triage

- The support agent reviews the ticket subject and description.
- The issue is categorized as **Technical** or **Account Management**, depending on the nature of the problem.
- The ticket is tagged as **login-issue**.
- Priority is set to **High**, and SLA timelines are assigned.
- The ticket status is updated to **Pending**.

3. Investigation

- The assigned agent reviews authentication logs and account status.
- Potential causes such as account lockouts, expired reset links, or security blocks are identified.
- If needed, the agent contacts the customer for verification or additional information.

4. Resolution

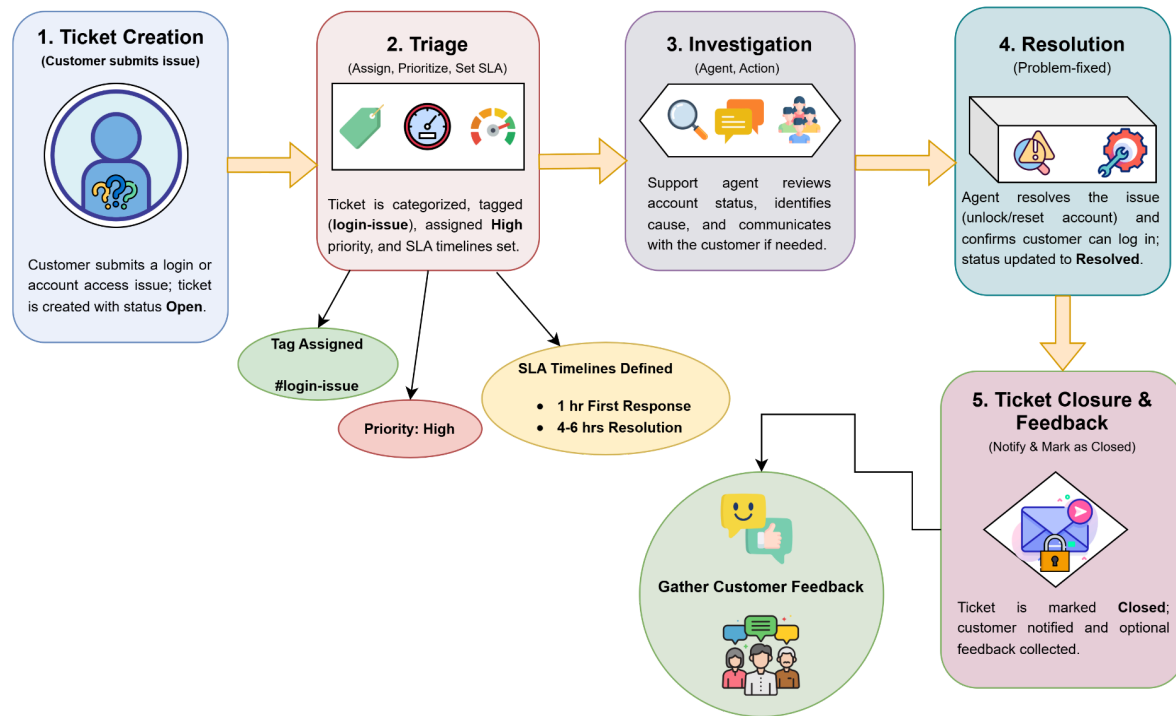
- Corrective actions are performed, such as unlocking the account or resetting credentials.
- The agent confirms that the customer can successfully log in.
- The ticket status is updated to **Resolved**.

5. Closure & Feedback

- After confirmation, the ticket is marked as **Closed**.
- A resolution summary is sent to the customer.

- Customer feedback may be collected to assess support quality.

Based on the analysis, our proposed “**customer support workflow diagram**” is drawn below:



Overall Flow Summary (One-Line)

Customer submits login issue → Ticket is triaged with tag, priority, and SLA → Agent investigates → Issue is resolved → Ticket is closed and feedback is gathered.

This simulated support flow demonstrates a structured and consistent approach to resolving high-impact recurring issues. By applying standardized triage rules, realistic SLA timelines, and clear resolution steps, support efficiency and customer experience can be significantly improved.

8. Key Insights & Observations

This section briefly summarizes the main insights gained from reviewing and categorizing the support tickets.

- Most customer problems fall into a few main areas.

Many tickets relate to login and account access, technical problems, and billing issues.

These areas cause the most difficulty for customers and have the biggest effect on their experience.

- Setting the right priority helps issues get resolved faster.

Problems that stop customers from using the platform, such as login or payment issues, are given higher priority. This helps the support team focus first on the most serious problems.

- Using clear and consistent tags makes issues easier to manage.

Applying the same tags to similar tickets helps group related problems together. This makes it easier to spot repeated issues and handle them in a more organized way.

- Clear time targets improve the support process.

Defining how quickly support should respond and resolve issues helps create a more reliable and predictable support experience for customers.

9. Learning Outcomes

During Week 1, we gained a clearer understanding of how customer support systems work in practice. We learned how to review support tickets carefully and classify them based on the information provided, using simple and consistent rules.

We also developed the ability to identify repeated customer problems and understand what they reveal about the product and service. Overall, this exercise helped us build a stronger habit of using data to guide decisions and improve customer support quality.

10. Conclusion

The Week 1 Ticket Mapping activity provided valuable hands-on exposure to customer support operations through the application of structured data analysis techniques. By systematically reviewing and categorizing customer support tickets using clearly defined criteria, we developed a strong understanding of how customer issues are identified, classified, prioritized, and resolved within a real-world support environment. This process highlighted the importance of consistency, accuracy, and structured decision-making in effective support operations.

The analysis enabled the identification of recurring customer issues, including login and account access problems, technical errors, and billing-related concerns. These recurring patterns revealed key areas that have a significant impact on customer experience and operational efficiency. Additionally, simulating an end-to-end support workflow for a high-impact recurring issue enhanced our understanding of ticket triage, priority assignment, SLA management, and resolution tracking. This exercise demonstrated how structured workflows and clearly defined response timelines contribute to faster issue resolution, improved service quality, and higher customer satisfaction.

End of Report
Thank You !

Customer Insights and Data Analytics Early Remote Internship

Excelerate

Sub-Group 2 : Team Charter

Team Members	Active Members: Sumaiya Tasnim (sumaiyaa.tasnim.18@gmail.com) Omotobora Alonge (toboraalonge20@gmail.com) Areeba Imtiaz (areebarebo09@gmail.com) Ajay Ahirwar (kgfgamingzone@gmail.com) Japheth Joseck (joseckjeff@gmail.com) Inactive Members: khaled lotfy (khaled.sherif.lotfy@gmail.com) Jennifer Mens (jennifermens718@gmail.com) Kahkshan Shahnaz (kahkshan6001@gmail.com) Jedida Waweru (wawerujedida@gmail.com)
Team Lead	Sumaiya Tasnim (sumaiyaa.tasnim.18@gmail.com)
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Team Members Roles and Responsibilities	Team Members Roles and Responsibilities: Sumaiya Tasnim (sumaiyaa.tasnim.18@gmail.com) - Team Lead, represents team to sponsor, via email and on calls, to minimize communication errors. Japheth Joseck (joseckjeff@gmail.com) - Project Manager, provides guidance and draws out insight from other team members, ensures that the project execution remains on track. Areeba Imtiaz (areebarebo09@gmail.com) - Project Scribe, responsible to taking meeting minutes and distributing notes/assignments. Can assist Team Lead in drafting emails and communication between sponsor and group. Omotobora Alonge (toboraalonge20@gmail.com) - Project Lead, responsible for holding the group accountable for meeting deadlines and ensures that the project deliverables are being met.
Mission, Vision Objectives & Core Values	Mission: To analyze customer support and service data to identify trends and generate actionable insights that enhance operational efficiency and customer satisfaction. Our mission focuses on monitoring key metrics such as ticket volume, response times, resolution rates, and SLA compliance through structured reporting, dashboards, and automated analytics, enabling data-driven improvements in support processes. Vision: To build a high-impact customer insights and data analytics function that transforms raw support data into meaningful intelligence. We aim to support customer-centric decision-making by improving service workflows, strengthening agent productivity, and mapping the customer journey to deliver scalable, efficient, and consistently positive customer experiences.

	Core Values: ● Integrity – We uphold transparency and accuracy in every task and report. ● Accountability – We take ownership of our responsibilities and deliver on time. ● Collaboration – We support one another through clear communication and teamwork. ● Excellence – We aim for continuous improvement and high-quality results. ● Responsiveness – We stay proactive and ensure timely communication across all channels. ● Growth Mindset – We view every challenge as a learning opportunity to refine our skills and strengthen our impact.
Internal Checks, Balances, and Reviews	● Team check-ins will occur daily or every 2 days unless cancelled in agreement with all team members. ● The tasks leading up to the final deliverable submission will be split between team members in an even manner. ● Team members are expected to help others with their work to ensure the completion and growth of other members. ● Daily or alternate-day check-in meetings will be used to ensure all team members are on track and ensure all doubts are getting clarified. ● Team lead and Project manager are responsible for leading meetings unless they are unable to attend and must notify so and delegate their responsibilities to someone else. ● Then, team members will work to set an agenda for the next meeting and the appropriate task deadlines. ● Project Scribe will keep a record of group discussions, everyone's tasks, and inactive members. They will provide the documentation after the meeting in Google Chat. ● The team will have a final review of each team member's contributions to reduce errors, and they will collaboratively create the final submission for the week.

Operations: • Assignments • Meetings • Communication Guidelines • Status Updates • Deadlines	Assignments: Tasks are distributed to members present during meetings. Absent members can be assigned tasks if they respond within 24 hours. Everyone should check the Task Tab regularly for updates and assigned roles. Must acknowledge assigned tasks promptly and meet deliverables. Meetings: Regular meetings every Wednesdays are required for alignment and updates. First absence is excused; consistent attendance afterward is expected. Must attend all remaining meetings and actively participate. Communication Guidelines: Open, respectful, and consistent communication across channels. Reply within 24 hours and check group chats every hour for updates. Stay responsive and maintain professionalism in all group communications. Status Updates: Progress must be visible and shared for transparency. Members provide regular updates in the chat or task tracker. Non-reporting may affect task reassignment or accountability evaluation. Deadlines: All deliverables must meet agreed timelines. For members, weekly deliverable deadline is on every Saturday at 11:00 PM IST. Submit outputs on or before the set deadline. Late submissions require valid reasons. As our all weekly submissions are on monday in Excelerate portal, so we can verify/improve our works on sunday by collaborating with each other.
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